

Kovidh Vardhan Singh Bhati

+91-9829862825 | kovidhvsb7724@gmail.com | linkedin.com/in/kovidhbhati | github.com/kvsb25

SUMMARY

Computer Science undergraduate at VIT (CGPA: 8.73/10) with hands-on experience building production-grade systems in TypeScript, C++, and JavaScript. Proven track record designing event-driven architectures, multi-agent AI pipelines, and real-time peer-to-peer applications. Seeking software engineering internships or entry-level roles in backend systems, cloud infrastructure, or AI-driven products.

EDUCATION

Vellore Institute Of Technology

B.Tech in Computer Science and Engineering | CGPA: 8.73/10.0

Sep 2023 – May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, OOP, Cloud Computing

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, C++, HTML, CSS
- **Databases:** SQL (MySQL), NoSQL (MongoDB, Redis)
- **Frameworks & Libraries:** Express.js, React.js, REST APIs, WebSockets, WebRTC, gRPC
- **Cloud & DevOps:** Docker, Git, GitHub Actions CI/CD, Apache Kafka, Postman, AWS
- **AI:** Generative AI, Retrieval-Augmented Generation (RAG), Multi-Agent AI (RAG, HITL) LangChain, LangGraph

PROJECTS

Morpheus – Agentic SRE | *TypeScript, LangGraph, LangChain, Gemini, Docker* | [GitHub](#)

- Built an event-driven, multi-agent incident-response system (LangGraph/LangChain, Gemini 3.5 Flash) automating a 4-phase RCA pipeline, with a ChromaDB RAG pipeline over runbooks and a structured Reviewer loop to cut hallucinations.
- Gated a whitelist-enforced remediation executor behind Slack HITL approval; integrated an LTOV observability stack (Loki, Tempo, OpenTelemetry, VictoriaMetrics) with SLI-based alerting routed via Alertmanager/vmalert to trigger the agent with zero pollin

Zen – C++ Web Micro-Framework | *C++, Winsock2, TCP/IP, HTTP* | [GitHub](#)

- Engineered a C++ web framework using Winsock2 with a multi-threaded connection pool for load distribution and RAII-driven ClientSessions for memory-safe, single-owner socket lifecycle management.
- Architected a 3-tier deterministic middleware pipeline (Default, Path, Error) with centralized routing, and built custom HTTP/URL parsers plus a priority-queue content negotiator exposing a fluent function-chaining API.

Face2Face – P2P Video Conferencing | *JavaScript, Express.js, WebRTC, WebSocket, Docker* | [GitHub](#)

- Built a private one-to-one video conferencing app with real-time chat and file sharing; implemented a WebSocket signaling server for WebRTC ICE candidate exchange enabling secure NAT-traversing peer-to-peer connections.
- Containerized the HTTP and WebSocket servers using Docker, ensuring consistent and isolated deployment environments across development and production.

EXPERIENCE

Backend Development Intern

Apr 2026 – Jun 2026

TEKNOSys

Remote

- Architected and deployed a RESTful authentication and role-based access control service in TypeScript (ExpressJS) with PostgreSQL and Prisma, implementing secure JWT session handling and fine-grained permission middleware.
- Integrated Redis-based caching and rate limiting alongside strict Zod payload validation, reducing API response latency and mitigating brute-force risk while keeping availability consistent during traffic spikes.

Frontend Development Intern

Jul 2025 – Aug 2025

StudyBase Pvt. Ltd.

Remote

- Engineered a greenfield React Address Book using react-window virtualization and custom state hooks, enabling smooth rendering and search across 500+ contacts.
- Integrated REST APIs via Axios, conducted endpoint testing with Postman, and followed Git feature-branch workflows, reducing bug resolution time by an estimated 20%.

CERTIFICATIONS

- **Microsoft Certified:** Azure Fundamentals (AZ-900)
- **IBM Certified:** AI Fundamentals

ACHIEVEMENTS

- Selected among top 1,610 teams out of 35,100+ in Flipkart Gridlock 2.0 hackathon (top 4.6%) for proposing a data-driven solution to Bangalore's traffic congestion.